8.1 INTRODUCTION

Based on technology, the global microbiome sequencing services market is segmented into sequencing by synthesis (SBS), sequencing by ligation (SBL), nanopore sequencing, and other technologies (semiconductor sequencing and Sanger sequencing, among other emerging technologies). The SBS segment accounted for the largest market share (~45%) in 2022, while the nanopore sequencing segment is estimated to grow at the highest CAGR (17.3%) during the forecast period.

TABLE 50 MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	99.59	113.38	129.01	146.93	167.48	191.59	219.92	253.56	14.5%
Sequencing by Ligation	28.18	31.96	36.24	41.13	46.71	53.24	60.89	69.95	14.1%
Nanopore Sequencing	11.29	13.20	15.42	18.01	21.05	24.67	29.00	34.21	17.3%
Other Technologies	80.97	91.63	103.64	117.32	132.92	151.11	172.40	197.53	13.8%
Total	**220.03**	**250.17**	**284.31**	**323.39**	**368.16**	**420.61**	**482.21**	**555.24**	**14.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

8.2 SEQUENCING BY SYNTHESIS (SBS)

8.2.1 SBS TECHNOLOGY ENABLES HIGHEST PRODUCTION OF BASE PAIRS IN SHORT TIME

The SBS technology uses four fluorescently labeled nucleotides to sequence millions of clusters on the flow cell surface in parallel. During every sequencing cycle, a single-labeled deoxyribonucleoside triphosphate (dNTP) is added to the nucleic acid chain. This nucleotide label serves as a terminator for polymerization. After dNTP incorporation, the dye is first imaged to identify the base, and then it is enzymatically cleaved to allow the incorporation of another nucleotide. Since all four reversible terminator-bound dNTPs are present during each sequencing cycle, natural competition minimizes incorporation bias. The result is true base-by-base sequencing that enables the most accurate data for a broad range of applications.

The SBS technology has a wide customer base owing to its well-established precision in sequencing as well as its broad application base. It has applications in microbial genetics and metagenomics and is mainly used in whole-genome sequencing, transcriptome analysis, targeted sequencing, and genome-wide protein-nucleic acid interaction analysis.

TABLE 51 MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	57.03	64.63	73.20	82.98	94.15	107.20	122.48	140.55	13.9%
Europe	30.83	35.22	40.20	45.94	52.54	60.30	69.44	80.32	14.8%
Asia Pacific	8.47	9.93	11.63	13.62	15.96	18.75	22.09	26.12	17.6%
Latin America	1.93	2.14	2.37	2.63	2.92	3.25	3.62	4.05	11.3%
Middle East and Africa	1.33	1.46	1.60	1.75	1.91	2.10	2.30	2.52	9.5%
Total	**99.59**	**113.38**	**129.01**	**146.93**	**167.48**	**191.59**	**219.92**	**253.56**	**14.5%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 52 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	51.68	58.62	66.45	75.39	85.60	97.54	111.54	128.09	14.0%
Canada	5.34	6.01	6.75	7.59	8.55	9.66	10.94	12.46	13.0%
Total	**57.03**	**64.63**	**73.20**	**82.98**	**94.15**	**107.20**	**122.48**	**140.55**	**13.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 53 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	6.68	7.65	8.77	10.04	11.52	13.26	15.32	17.77	15.2%
UK	5.97	6.87	7.91	9.10	10.48	12.12	14.06	16.38	15.7%
France	4.72	5.39	6.15	7.03	8.05	9.24	10.64	12.31	14.9%
Italy	1.33	1.51	1.72	1.96	2.24	2.56	2.94	3.39	14.5%
Spain	1.02	1.16	1.32	1.50	1.71	1.95	2.24	2.58	14.3%
Rest of Europe	11.11	12.62	14.34	16.30	18.54	21.16	24.24	27.89	14.2%
Total	**30.83**	**35.22**	**40.20**	**45.94**	**52.54**	**60.30**	**69.44**	**80.32**	**14.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis